ComputerVision Counter — Quick Start

This short guide reflects the current behavior: **AOI toggle respected**, **clean overlay summary**, and **immediate abort**.

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1) Install once

```
python bootstrap_env.py
```

Installs dependencies into a local `./pkgs` folder (portable; no system pollution).

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2) Launch

```
python start_app.py
```

Choose **Input** and **Model** via the GUI (not prefilled). Default output is `./output/`.

> **Supported models:** Ultralytics `.pt` only. **ONNX is disabled in this release** and will arrive in the next release.

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3) Load data & model

1. **Browse… (Input)** → pick a folder or select files.
2. **Browse… (Model)** → choose a YOLO `.pt` file. Class names auto-load.
3. **Select classes** (at least one).

Get a test model (YOLO11n)
We do **not** bundle third-party weights. To try the app quickly, fetch a small official model from Ultralytics: - Ultralytics Detect task: https://docs.ultralytics.com/tasks/detect/ - YOLO11 models overview: https://docs.ultralytics.com/models/yolo11/

> Tip: choose **YOLO11n** (nano) for the fastest test.

More model sources
- **Ultralytics Docs** (links above) — official `.pt` weights.
- **Ultralytics on Hugging Face** — search for `yolo11n.pt` and related variants.
- **Model zoos** (varied licenses): Roboflow Universe, Hugging Face Hub. Always verify the license of any third-party model before use.

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4) (Optional) Draw AOIs

Open **AOI Editor** and draw named polygons.

- Finish polygon: **Ctrl + Enter**
- Undo vertex: **Ctrl + Backspace**
- AOIs auto-save to `input/aoi/*.json` + `input/aoi_masks/*.png`.

AOI modes

- **Centroid** (default)
- **Box overlap** with `aoi_box_frac` (e.g., 0.20 = 20%).

Important: The main **Use AOI** toggle is authoritative — if **OFF**, AOIs (even if present on disk) are **ignored**.

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5) Choose quality

Use the **Fast / Balanced / Ultra** presets, or open **Advanced** for tiling, thresholds (strict `conf`), WBF, seam de-dup, and overlay controls.

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6) Run & abort

Click **Start**. Progress is tile-based with ETA.

Click **Abort** to stop immediately; the label changes from **Aborting…** to **Aborted.** as soon as the worker finishes the current tile.

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7) Outputs

All in `./output/` (non-destructive filenames):

- `annotated/<image>_annotated.jpg`
- **With AOIs** → `Total …   AOI1 …   AOI2 …`
- **Without AOIs** → `Total … (per-class breakdown)`
- `results_per_image.csv` & `results_totals.json`
- `detections_full.csv` (kept detections; includes AOI name when AOI is ON)
- `gis/<image>__detections_point.csv` and `gis/<image>__detections_box.csv` for georeferenced images

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8) Tips

- Ensure **at least one class** is selected.
- Larger `tile` + higher `overlap` → better quality, slower.

- If overlays and CSV disagree slightly, remember CSV uses raw floats and strict `>` on `conf`.

🔭Roadmap (next release)

- **ONNX runtime support** (first official release).
- **Segmentation models** support.
- **Classification models** support.
- Further improvements to AOI handling and overlay summaries.

Licensing note

We don’t include third-party model weights in the app package. Ultralytics YOLO models are provided under **AGPL-3.0** (or a commercial **Enterprise License**). Review the model’s license before **redistributing** any weights or using them in commercial settings.